THE ORIGIN OF THE EARTH

BY JAIPRAKASH RAU

- Earth is the third planet of the solar system as counted from the sun. The Earth also came into existence almost the same time as the other members of the solar system, i.e. around 4.5 billion years ago.

- Gas is a state of matter that has no fixed shape and no fixed volume. Gases have lower density than other states of matter, such as solids and liquids. There is a great deal of empty space between particles, which have a lot of kinetic energy

- In this gaseous state of matter a disturbance occurred and as a result condensation was started.

- As a result of condensation the latent heat was released and it increased the temperature from 500 degrees to 5000 degrees.

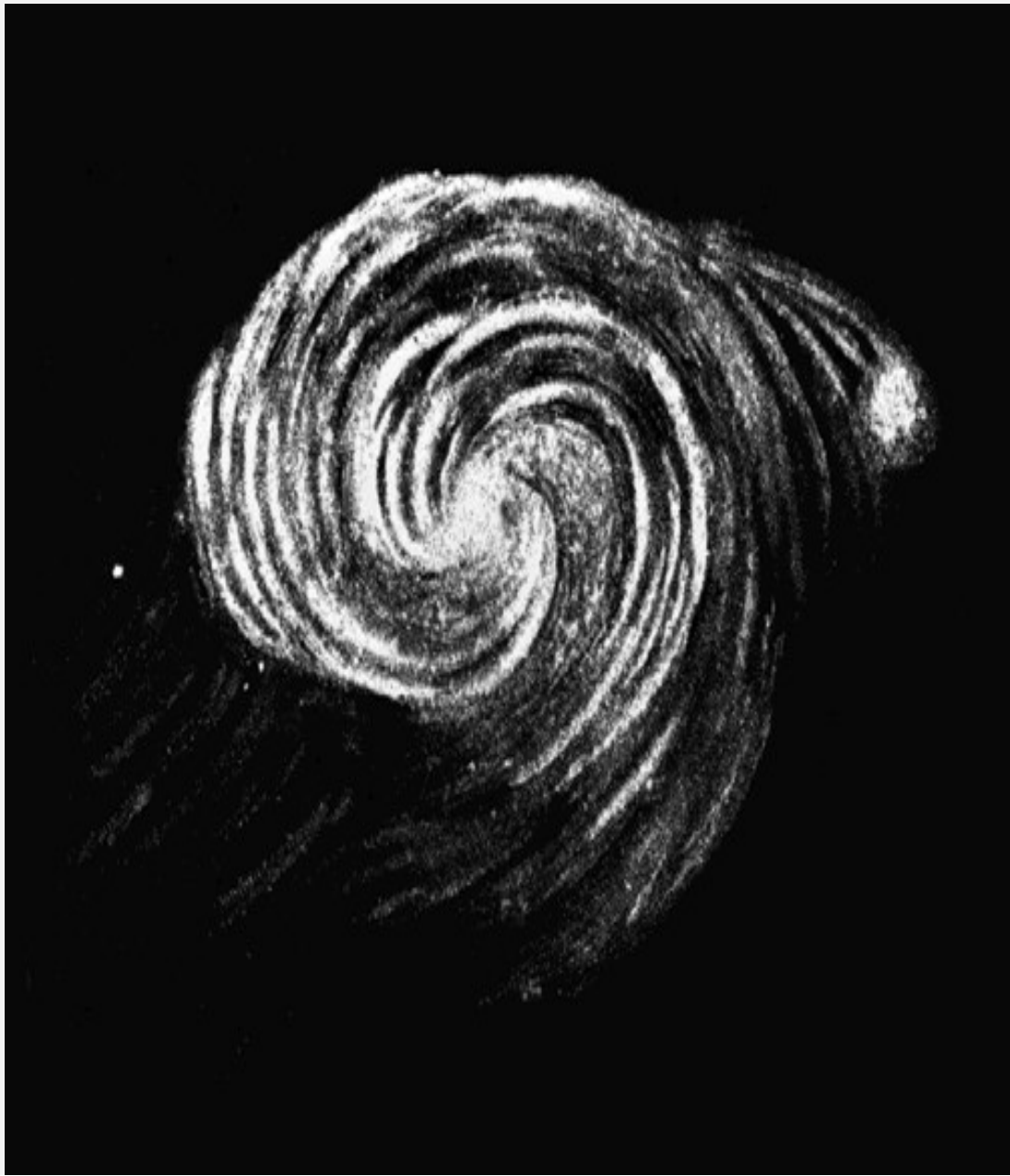
- The disturbance in the universe and the condensation has been a subject to great discussion and speculation.

- As such, numerous theories have been advanced with regard to the composition, rotation, and condensation of the spiral nebulae.

- Although some believed that the nebulae were composed f solid meteorites but this is no longer subscribed and all the authorities agree on one point that the spiral nebulae were a gaseous mass.

Spiral Nebulae are objects that look like spiral shaped clouds which later were found to be galaxies themselves lying outside our milky way galaxy.

Evidence of Spiral structure in the distant galaxies was first noticed by William Parsons in 1845.



GEOCNTRIC THEORY

- The earliest systematic theory of the Earth was the "Geocentric (geo = earth) theory".

- According to this theory, the earth was the unmoving centre of the universe, round which the sun and the stars and all other heavenly bodies revolved.

- The final formulation of the theory was made by CLAUDIUS PTOLEMY, a Greek astronomer of Alexandria.

HELIOCENTIC THEORY

- The Heliocentric Theory (Helios = sun) was first advanced by NICOLAU KOPPERNIGK (1473 – 1543), a Polish astronomer, better known by his Latinised name COPERNICUS.

- He set out the theory that the sun was the centre of the universe and the earth and other planets revolved round it.

MODERN THEORIES ON THE ORIGIN OF THE EARTH

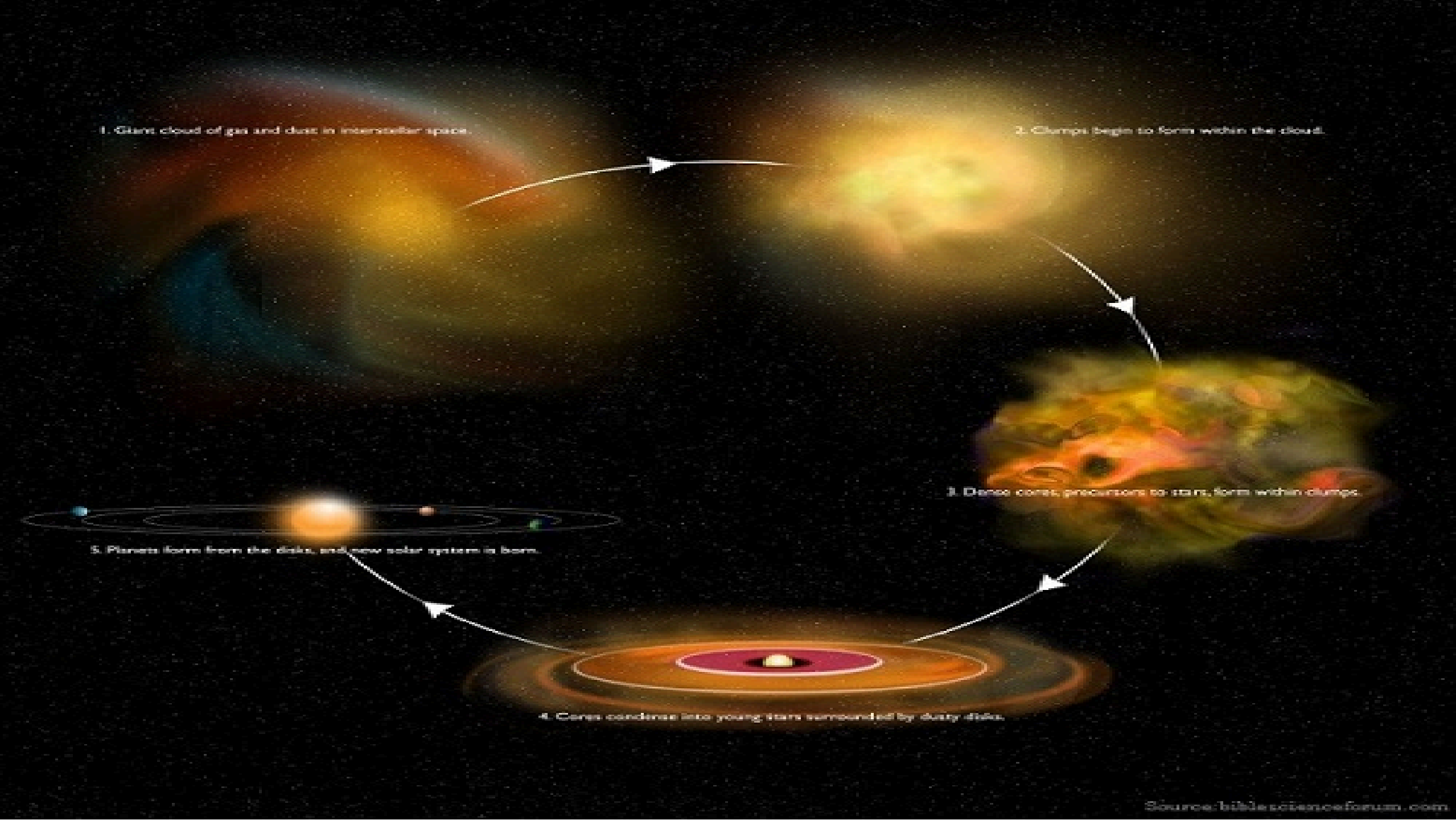
- **BUFFON'S HYPOTHESIS**
- Georges de Buffon, a French scientist, was the first to give a scientific explanation regarding the origin of the earth in 1745.

- According to Buffon, a comet came very close to the sun and collided with it due to gravitational pull.

- As a result of this collision, a lot of material was ejected from the Sun and got separated from it.

- The separated matter formed planets and asteroids after condensation due to cooling.

- The bigger masses became planets and the smaller masses became asteroids. Some of the lighter matter disappeared in the space permanently.

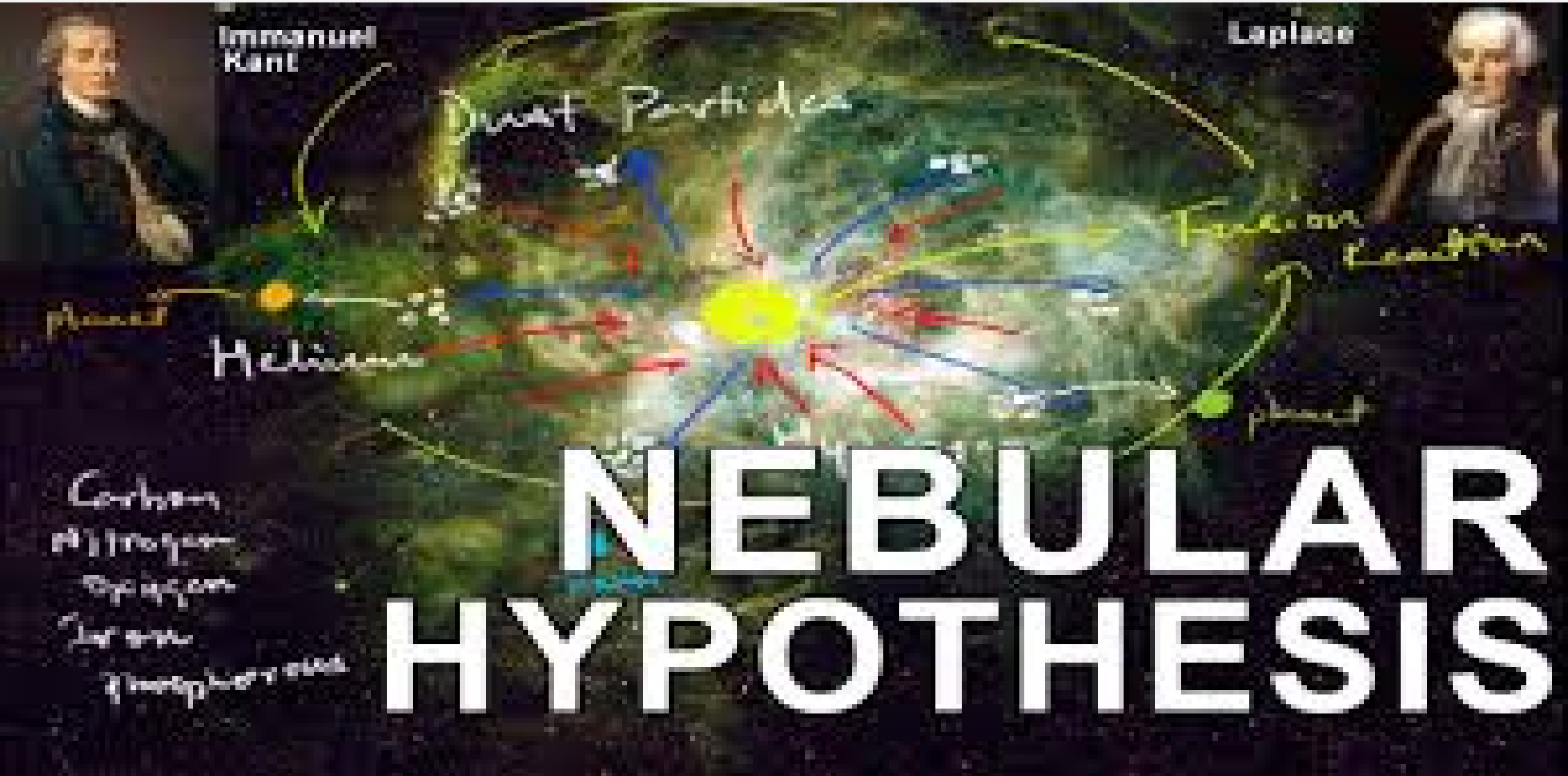


Source: biblescienceforum.com

NEBULAR HYPOTHESIS

- The Solar Nebula Theory postulates that the Solar System is formed by the transition from an undifferentiated cloud to a star system and then completed with planets and moons around 100 million years ago. This theory states that the Solar System was formed in 4.6 billion years ago by condensation

- Nebular Hypothesis considered that the planets were formed out of a cloud of material associated with a youthful sun, which was slowly rotating. Later in 1950, Otto Schmidt in Russia and Carl Weizascar in Germany somewhat revised the ‘nebular hypothesis’, though differing in details.
  1. They considered that the sun was surrounded by solar nebula containing mostly the hydrogen and helium along with what may be termed as dust.
  2. The friction and collision of particles led to formation of a disk-shaped cloud and the planets were formed through the process of accretion.


Immanuel Kant
Laplace
NEBULAR HYPOTHESIS

FORMATION OF THE SOLAR SYSTEM

- Nebular Theory of Laplace (1796) tried to explain the formation of the solar system. But it had many drawbacks as the theory was based on scientifically erroneous assumptions.
- But one assumption it got right was that the solar system was born from a giant gas of dust called as nebula.
- A giant interstellar cloud known as the solar nebula (a vast, swirling cloud of gas and dust) gave birth to our solar system and everything in it.
- The nebula started its collapse and core formation some 5-5.6 billion years ago and the Sun and the planets were formed about 4.6 billion years ago.

FORMATION OF THE SUN

- The nebula began to collapse (gravitational collapse) in on itself after becoming gravitationally unstable.
- This was possibly because of a nearby supernova sending shock waves rippling through space.
- Gravity then caused dust and gas to coalesce to the centre of the nebular cloud.
- As more matter got pulled in, the centre got denser and hotter, increasing the gravity and pulling even more dust inwards causing a snowball effect.
- About 99.9% of the material fell into the centre and became the protosun (no sunlight yet).
- Once the centre of the cloud became hot enough it triggered nuclear fusion, and the Sun was born.